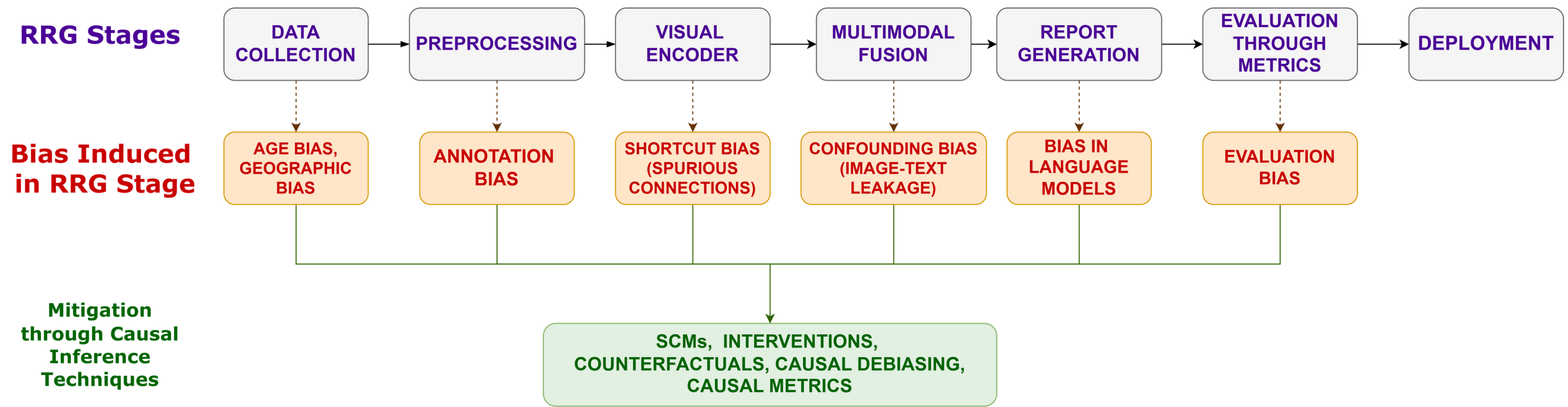


Introduction

- **Radiology:** extremely important diagnostic tool saving millions of lives per year.
- But there is a *global shortage of radiologists*.
- **Automated RRG with Deep learning:**
 - Encode images and large language models (LLMs) to generate text.
 - Incorporate knowledge graph to improve accuracy.
 - Final check with radiologists.
- **Issue:** Such systems are vulnerable to biases stemming from data imbalances, cognitive biases in human annotations, and limitations of LLMs.
- **Potential Solution:** *Causal Inference* for identifying true cause-effect relationships in reports and separating them from spurious correlations in observational data.

Biases in RRG

- **Bias in Radiology Practice and Reporting:**
 - **Alliterative Bias:** relying heavily on prior reports, perpetuating errors through a chain.
 - **Framing Bias:** Interpretation skewed by limited or misleading clinical history provided by the referring clinician.
 - **Availability Bias:** Easily recalled or recent cases are overestimated.
 - **Anchoring Bias:** Sticking to the first diagnosis, even when new evidence proves it wrong.
 - **Hindsight bias:** Overestimating predictability after knowing outcomes.
 - **Blind Spot bias:** Identifying others' errors while failing to recognize one's own.
 - **Scout Neglect bias:** Overlooking findings in preliminary scout images because pathology is not expected there.
- **Bias in Medical Imaging:**
 - **Sampling Bias:** Dataset fails to represent the full diversity of patient populations. Results in higher error rates for underrepresented subgroups like marginalized ethnicities.
 - **Annotation Bias:** Subjective errors and variability in how clinicians label images. Often leads to over-focus on malignant lesions while under-annotating benign ones.
 - **Propagation Bias:** Data quality issues (missing metadata, transcription errors) that multiply throughout the AI training pipeline and amplify in the final model.
 - **Acquisition Bias:** Systematic differences due to scanner types or imaging protocols. Models trained on high-res scans often fail on routine clinical imaging.
- **Bias due to LLMs:**
 - **Hallucinations:** Describing a condition not present in the image, i.e., may generate findings with content that cannot be directly linked to the input information, a serious danger in clinical use.
 - **Dataset imbalances and spurious correlations:** LLMs may learn spurious correlations between clinical findings and patient demographics or co-occurring diseases that do not reflect true causal relationships
 - **Sociodemographic bias:** LLM outputs can vary by patient demographics. Studies found that GPT-3.5 and GPT-4 can use race-based information while answering questions.



RRG Pipeline with bias induced in each step.

Causal Modeling of RRG

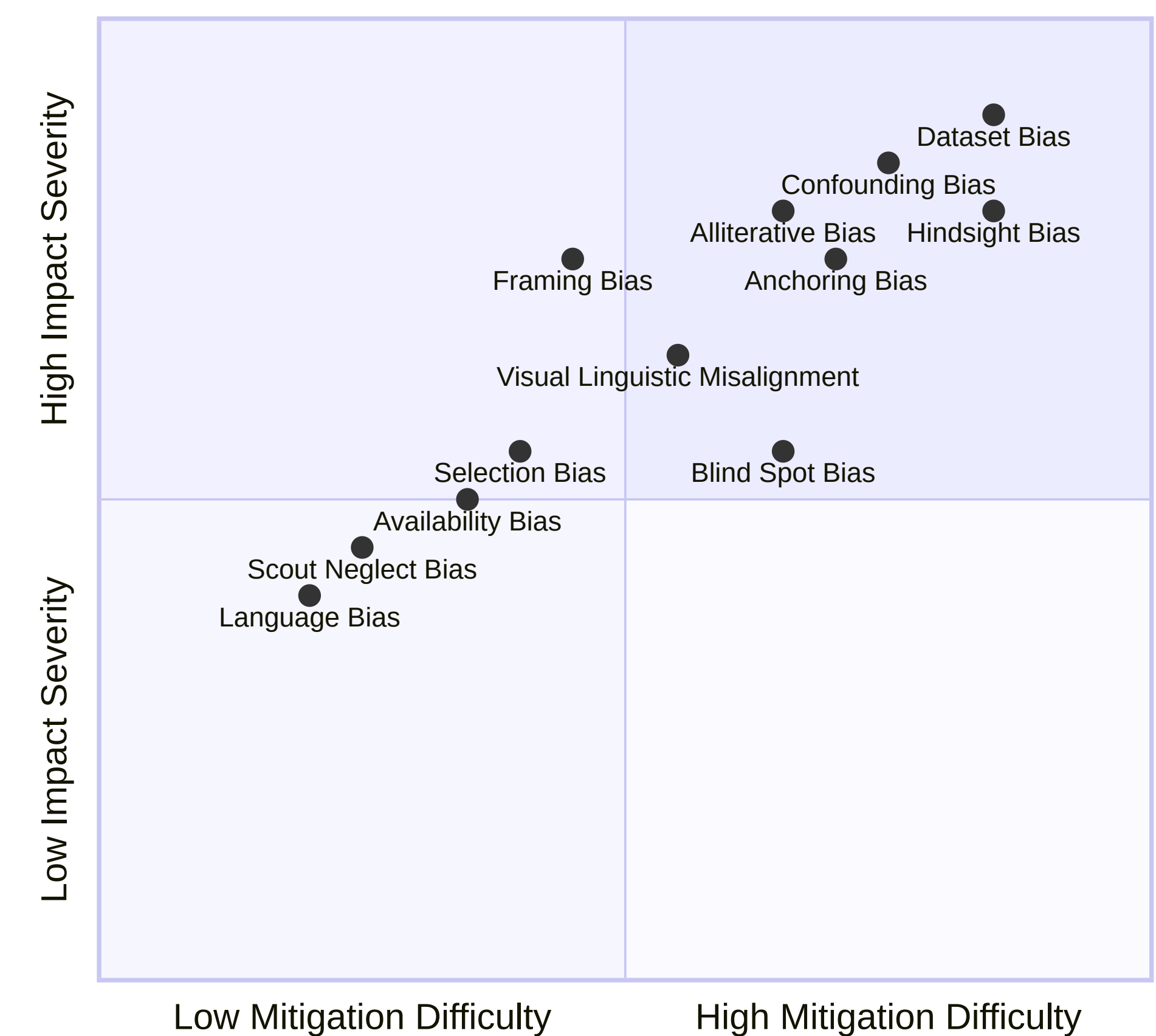
- RRG can be framed as a causal process involving multiple variables such as the medical image X , clinical context C , radiologist's interpretation I , and the final report text R .
- It can be modeled using an SCM defined as a tuple $\mathcal{M} = \langle U, V, F \rangle$, where U is a set of exogenous variables (unobserved noise), $V = \{X, C, I, R\}$ is a set of endogenous variables, and the set of causal mechanisms denoted by $F = \{f_X, f_C, f_I, f_R\}$.
- Causal Mechanism:
$$\begin{cases} X = f_X(U_X) \\ C = f_C(U_C) \\ I = f_I(X, C, U_I) \\ R = f_R(I, U_R) \end{cases}$$
- f_I captures how the image features and the clinical context influence the radiologist's interpretation, while f_R models how the interpretation generates the textual report.

Mitigation Strategies

We broadly categorize various interventions against bias in RRG:

1. **Data-level intervention**
 - improving the quality and representativeness of training data to reduce confounding and selection biases inherent in radiology datasets.
 - adjusting the dataset to equalize representation.
 - use of causal sampling, re-annotation and domain knowledge helps.
2. **Model-level intervention**
 - Incorporates causal principles directly into architecture and training objectives.
 - **Deconfounding:** Mitigating cross-modal confounders via causal frontdoor interventions.
 - **Debiased Prompting:** Fine-tuning LLM decoders with causal prompting to ensure pathological grounding.
3. **Evaluation-level intervention**
 - Changes how fairness is measured and enforced during testing and deployment without changing the model.
 - **Counterfactuals simulation** in protected attributes to reveal hidden biases.
 - **Interpretability:** Identifying residual biases to help clinicians interpret decisions causally.
 - **Targeted Audits:** Essential radiologist review focusing on hallucinations and representation failures.

RRG Bias Matrix: Severity vs Mitigation Difficulty



Bias in Medical Imaging Datasets

Organ	Datasets	Geographic Bias	Age Bias
Chest	MIMIC-CXR (Johnson et al., 2019)	United States (Boston)	Predominantly elderly ICU patients
	IU-Xray (Demner-Fushman et al., 2015)	United States (Indiana University)	Middle-aged (mean age ≈ 49 years)
	CheXpert (Irvin et al., 2019)	United States (Stanford University)	Middle-aged adults
	PadChest (Bustos et al., 2020)	Spain	Elderly patients
Spine	OAI (Peterfy et al., 2008), RSNA Spine (Flanders et al., 2022)	Germany, United States	Degenerative cases overrepresented
	LiTS (Bilic et al., 2023)	United States	
Abdomen	KiTS (Heller et al., 2019)	United States (University of Minnesota Medical Center)	Middle-aged adults
	CHAOS (Kavur et al., 2021)	China	
Brain	BraTS (Menze et al., 2015), CQ500 (Chilamkurthy et al., 2018)	India, United States, China	Adult glioma cases (age ≥ 35 years)

Table 2: Summary of age and geographic biases in commonly used medical imaging datasets.

Bias in Evaluation Metrics

- Conventional Metrics (e.g., BLEU, ROUGE, METEOR, CIDEr) measure textual overlap but are agnostic to causal structure. As a result, these metrics cannot detect or correct biases arising from confounding, mediation, or selection bias.
- Motivates the development of causal-aware metrics presented below:

Metrics	Causal Variable Evaluated	Causal Estimand	Bias Type Detected
CheXpertF1 (Irvin et al., 2019)	Clinical findings (disease presence)	$P(F \text{do}(X))$	Confounding, annotation bias
RadGraph F1 (Jain et al., 2021)	Entities & relations	$P(E, R \text{do}(X))$	Mediation
Demographic Performance Gap (Obermeyer et al., 2019)	Group-conditioned findings	$\mathbb{E}[Y \text{do}(D=g)]$	Demographic confounding
Calibration Error (ECE/Brier) (Guo et al., 2017; Subbaswamy and Saria, 2019)	Model confidence vs correctness	$P(Y \text{do}(X))$	Selection bias, dataset shift
Counterfactual Consistency (Pearl et al., 2016)	Stability under irrelevant changes	$Y_{A=a} = Y_{A=a'}$	Shortcut learning
Invariant Risk / Stability Score (Arjovsky et al., 2019; Gulrajani and Lopez-Paz, 2021; Magliacane et al., 2018)	Performance across environments	$Y \perp E \text{do}(X)$	Dataset shift
Transportability Error (Castro et al., 2020; Bareinboim and Pearl, 2013)	Cross-dataset correctness	$P(Y \text{do}(X))$	Institutional bias

Table 3: Causal-aware metrics applicable in RRG. F is findings, X is input, E, R are entity relationships, Y is outcome, A is variable name, and $\mathbb{E}(\cdot)$ is expectation.

Conclusion and Future Directions

- Causal Inference can act a key tool for tackling bias in RRG and improving the explainability and performance of current RRG systems; intervention of the radiologist isnecessary.
- The analysis presented in the work motivates various future directions:
 - Increasing demographics in RRG datasets
 - Better causal-aware metrics for RRG
 - Application of causal-inference to similar areas such as medical report summarization.